

Bayesian Linear Travel-Time Tomography: A Controlled Numerical Study of Reconstruction and Uncertainty Calibration

Abstract

We study a finite-dimensional Bayesian inverse problem motivated by crosshole seismic travel-time tomography. The unknown subsurface slowness field is discretized on a uniform square grid inside a square domain, while travel-time measurements are modeled as line integrals along known straight rays. This gives a linear forward model

$$\mathbf{d} = A\mathbf{s} + \boldsymbol{\varepsilon}.$$

We place a Gaussian prior on the discretized slowness field using a squared-exponential covariance, and assume Gaussian observation noise. The resulting posterior distribution is available in closed form.

The project has two goals. First, we construct and validate the complete forward and Bayesian inversion pipeline. Second, we investigate uncertainty calibration in a controlled synthetic setting. In particular, we distinguish between a correctly specified Bayesian generative model, where the true field is drawn from the assumed prior, and a fixed physical truth that is not itself regarded as a draw from the prior — including a case where that fixed truth is well matched to the prior’s assumed smoothness, and a case where it is deliberately mismatched.

The resulting study is intended as a small, reproducible computational project rather than as a claim of a new tomography algorithm. Its main purpose is to demonstrate, in a fully closed-form setting, when a mathematically well-defined posterior distribution is — and is not — a quantitatively meaningful description of physical uncertainty.

1 Problem statement

We consider a two-dimensional square domain

$$\Omega = [0, W] \times [0, W].$$

The unknown medium is described by its slowness field

$$s(\mathbf{x}) = \frac{1}{v(\mathbf{x})}, \quad \mathbf{x} \in \Omega,$$

where $v(\mathbf{x})$ denotes wave speed.

Under the straight-ray approximation, the travel time associated with a ray $L_i \subset \Omega$ is

$$t_i = \int_{L_i} s(\mathbf{x}) d\ell.$$

We place sources along the left boundary of the domain and receivers along the right boundary. Every source is connected to every receiver by a straight line segment. The resulting collection of rays provides the acquisition geometry.

The straight-ray approximation is adopted deliberately. In a fully physical treatment, rays may depend on the slowness field through Fermat’s principle, resulting in a nonlinear forward problem. Here the ray paths are fixed in advance so that the inverse problem remains linear.

2 Discretization

Partition Ω into a uniform $n \times n$ grid of square cells, giving n^2 cells total. Let $p = n^2$ denote this number of cells — equivalently, the number of unknown slowness values. The grid is uniform in both directions: cell width equals cell height. This eliminates cell aspect ratio as a variable, so that any anisotropy observed in the reconstruction or in posterior uncertainty is attributable only to the acquisition geometry and the prior.

We approximate the slowness field by a piecewise constant function:

$$s(\mathbf{x}) \approx s_j \quad \text{inside cell } j.$$

For ray i , let A_{ij} denote the length of the intersection of ray L_i with cell j . The discretized travel time is therefore

$$t_i = \sum_{j=1}^p A_{ij} s_j.$$

Writing $\mathbf{s} = (s_1, \dots, s_p)^\top$ and collecting the travel times into a vector $\mathbf{t} = (t_1, \dots, t_m)^\top$, we obtain the linear forward model

$$\mathbf{t} = A\mathbf{s}, \quad A \in \mathbb{R}^{m \times p}.$$

The matrix A depends only on the acquisition geometry and the discretization. It is therefore constructed once before the inverse problem is solved, and should be validated directly: for every ray i , $\sum_j A_{ij}$ must equal the total length of that ray, up to numerical precision (see Section 7).

3 Observation model

The measured travel times are modeled as

$$\mathbf{d} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma_d).$$

For the initial experiment we use independent, identically distributed Gaussian timing errors:

$$\Sigma_d = \sigma^2 I_m.$$

Thus $\mathbf{d} \mid \mathbf{s} \sim \mathcal{N}(A\mathbf{s}, \Sigma_d)$. The parameter σ controls the magnitude of measurement noise.

4 Gaussian prior

The discretized slowness vector is assigned a Gaussian prior

$$\mathbf{s} \sim \mathcal{N}(\mathbf{s}_0, C_s).$$

The prior mean is chosen to represent a homogeneous background medium, $\mathbf{s}_0 = s_{\text{bg}} \mathbf{1}$.

For the primary experiment we use a squared-exponential covariance. Let $\mathbf{x}_j$ denote the center of cell j . Define

$$(C_s)_{jk} = \tau^2 \exp\left(-\frac{\|\mathbf{x}_j - \mathbf{x}_k\|^2}{2\ell^2}\right) + \epsilon \delta_{jk},$$

where:

- τ^2 controls the marginal variance of the prior;
- ℓ is the spatial correlation length;
- ϵ is a small numerical jitter term added to the diagonal (e.g. $\epsilon \sim 10^{-6}\tau^2$).

Numerical stability note

The squared-exponential covariance is positive definite in exact arithmetic for any set of distinct grid points, but in floating-point practice it is frequently *numerically* near-singular — particularly at fine grid spacing or large ℓ , where many rows of C_s become nearly identical. Without the jitter term ϵ , Cholesky factorization or direct inversion of C_s (or of C_s^{-1} , as required by the posterior update in Section 5) will intermittently fail or produce badly conditioned results. The jitter term is standard practice in Gaussian process computation and should be treated as a required part of the implementation, not an optional safeguard.

The length scale ℓ controls the smoothness assumed by the inversion. Small ℓ permits relatively rapid spatial variation, whereas large ℓ strongly favors smooth fields.

5 Posterior distribution

Because the likelihood and prior are both Gaussian and the forward map is linear, the posterior distribution is Gaussian:

$$\mathbf{s} \mid \mathbf{d} \sim \mathcal{N}(\mathbf{s}_{\text{post}}, C_{\text{post}}).$$

The posterior covariance is

$$C_{\text{post}} = \left(A^\top \Sigma_d^{-1} A + C_s^{-1} \right)^{-1},$$

and the posterior mean is

$$\mathbf{s}_{\text{post}} = C_{\text{post}} \left(A^\top \Sigma_d^{-1} \mathbf{d} + C_s^{-1} \mathbf{s}_0 \right).$$

For the independent-noise model $\Sigma_d = \sigma^2 I$, these become

$$C_{\text{post}} = \left(\frac{1}{\sigma^2} A^\top A + C_s^{-1} \right)^{-1}, \quad \mathbf{s}_{\text{post}} = C_{\text{post}} \left(\frac{1}{\sigma^2} A^\top \mathbf{d} + C_s^{-1} \mathbf{s}_0 \right).$$

Thus the entire inverse problem can be solved without iterative optimization or Monte Carlo sampling. In implementation, both quantities should be computed via a Cholesky factorization of $(A^\top \Sigma_d^{-1} A + C_s^{-1})$ and triangular solves, rather than an explicit matrix inverse, for numerical stability.

6 What is being investigated?

The central computational question is:

Under what generative conditions does the posterior covariance of a Bayesian linear travel-time tomography model provide a quantitatively accurate description of repeated reconstruction uncertainty?

This question is investigated in a controlled synthetic setting, in which the true slowness field is known, by comparing two regimes: a correctly specified Bayesian generative model, in which the true field is itself drawn from the assumed prior (Experiment II), against a fixed physical truth that is not itself regarded as a draw from the prior (Experiment III). The fixed-truth comparison is run twice — once with a truth that is well matched to the prior’s assumed smoothness, and once with a truth that is deliberately mismatched — so that the resulting difference in calibration behavior is informative rather than a foregone conclusion.

7 Experiment I: construction and validation

The first stage is simply to construct the complete tomography pipeline.

Step 1: define the domain

Choose $W > 0$ and construct a uniform $n \times n$ grid of square cells. For an initial implementation, $n = 20$ is sufficient.

Step 2: define sources and receivers

Place N_s sources uniformly along the left boundary $x = 0$ and N_r receivers uniformly along the right boundary $x = W$. Every source is connected to every receiver, giving $m = N_s N_r$ rays.

Step 3: construct the sensitivity matrix

For each source–receiver pair, construct the corresponding straight line segment. For every grid cell, compute the length of the ray segment contained in that cell. These intersection lengths form the entries of A : $A_{ij} = \text{length}(L_i \cap \text{cell } j)$. As a basic validation, verify numerically that for every ray, $\sum_j A_{ij}$ equals the total length of that ray, up to numerical precision.

Step 4: define a synthetic truth

For the first reconstruction experiment, choose a smooth deterministic slowness field. One convenient choice is a homogeneous background plus a localized Gaussian anomaly:

$$s_{\text{true}}(x, y) = s_{\text{bg}} + \delta s \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2r^2}\right).$$

The corresponding cell values define $\mathbf{s}_{\text{true}} \in \mathbb{R}^p$. This deterministic truth is useful for directly measuring reconstruction error, and is used again in Experiment III below.

Step 5: generate synthetic data

Compute noiseless travel times $\mathbf{t}_{\text{true}} = A\mathbf{s}_{\text{true}}$. Then draw $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I)$ and form $\mathbf{d} = \mathbf{t}_{\text{true}} + \boldsymbol{\varepsilon}$.

Step 6: compute the posterior

Given $\mathbf{d}, A, \mathbf{s}_0, C_s, \sigma$, compute $\mathbf{s}_{\text{post}}$ and C_{post} . The main visualization should show: (1) the true field $\mathbf{s}_{\text{true}}$; (2) the posterior mean $\mathbf{s}_{\text{post}}$; (3) the posterior standard deviation $\sqrt{\text{diag}(C_{\text{post}})}$.

This establishes that the forward model and inverse calculation are working before any calibration experiment is attempted.

8 Experiment II: Bayesian calibration under the correctly specified model

The second experiment provides a clean statistical benchmark.

Instead of fixing $\mathbf{s}_{\text{true}}$, generate the true field from the same prior used by the inversion: $\mathbf{s}_{\text{true}} \sim \mathcal{N}(\mathbf{s}_0, C_s)$. Conditional on this realization, generate observations according to $\mathbf{d} \mid \mathbf{s}_{\text{true}} \sim \mathcal{N}(A\mathbf{s}_{\text{true}}, \Sigma_d)$. The inference model uses exactly the same C_s and Σ_d . This is the correctly specified Bayesian generative model.

For each independent realization:

1. draw $\mathbf{s}_{\text{true}} \sim \mathcal{N}(\mathbf{s}_0, C_s)$;
2. generate $\mathbf{d} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}$;
3. compute $\mathbf{s}_{\text{post}}$ and C_{post} ;
4. compute standardized errors;
5. compute the global Mahalanobis error.

For each cell define

$$z_j = \frac{s_{\text{true},j} - s_{\text{post},j}}{\sqrt{(C_{\text{post}})_{jj}}}.$$

Under the correctly specified Gaussian model, the posterior error conditional on the observed data satisfies $\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}} \mid \mathbf{d} \sim \mathcal{N}(0, C_{\text{post}})$. Consequently,

$$Q = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})^\top C_{\text{post}}^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})$$

satisfies $Q \sim \chi_p^2$. This provides an exact calibration benchmark: because the generative model and the inference model agree by construction, this experiment tests software correctness rather than any deeper statistical claim, and is a simulation-based-calibration-style check (cf. Talts et al., 2018), rather than a full implementation of that methodology. If Q does not follow χ_p^2 here, the bug is in the implementation, not in the statistics.

The numerical experiment should compare the empirical distribution of Q over many independent realizations with the theoretical χ_p^2 distribution.

9 Experiment III: deterministic truth and repeated observation noise

The previous experiment tests calibration under the complete Bayesian generative model, and by construction cannot fail unless the implementation is wrong. A different and practically important experiment is to hold the physical truth fixed and ask whether the posterior is still a trustworthy description of uncertainty — this is now a genuine statistical question, since nothing guarantees the answer is yes.

Choose a deterministic $\mathbf{s}_{\text{true}}$ and generate independent observations

$$\mathbf{d}^{(k)} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}^{(k)}, \quad \boldsymbol{\varepsilon}^{(k)} \sim \mathcal{N}(0, \sigma^2 I).$$

For each realization k , compute the corresponding posterior. This experiment answers a different question:

How well does the Bayesian posterior describe uncertainty around a particular fixed physical model when that model is not itself regarded as a random draw from the prior?

The same statistics can be collected:

$$z_j^{(k)} = \frac{s_{\text{true},j} - s_{\text{post},j}^{(k)}}{\sqrt{(C_{\text{post}}^{(k)})_{jj}}}, \quad Q^{(k)} = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}^{(k)})^\top (C_{\text{post}}^{(k)})^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}^{(k)}).$$

Importantly, there is no general reason for $Q^{(k)}$ to follow χ_p^2 in this fixed-truth experiment. The result depends on how the deterministic truth relates to the assumed prior.

Why $Q^{(k)}$ need not resemble χ_p^2 , even for a well-matched truth

The reason follows directly from the closed-form posterior update. Writing the posterior-mean error under a fixed truth using the update from Section 5,

$$\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}} = (I - KA)(\mathbf{s}_{\text{true}} - \mathbf{s}_0) - K\boldsymbol{\varepsilon}, \quad K = C_s A^\top (AC_s A^\top + \Sigma_d)^{-1},$$

splits the error into a *deterministic bias* $\mathbf{s}_b = (I - KA)(\mathbf{s}_{\text{true}} - \mathbf{s}_0)$ — fixed across realizations, since $\mathbf{s}_{\text{true}}$ does not vary — and a *noise-driven term* $-K\boldsymbol{\varepsilon}$, the only random part. The corresponding repeated-noise sampling covariance of $\mathbf{s}_{\text{post}}$ is

$$V_{\text{post}} = C_{\text{post}} A^\top \Sigma_d^{-1} A C_{\text{post}},$$

which is generally *not* equal to C_{post} itself: C_{post} also accounts for prior uncertainty about which $\mathbf{s}_{\text{true}}$ might hold, a component that is simply absent once $\mathbf{s}_{\text{true}}$ is fixed. Taking the expectation of $Q^{(k)}$ over repeated noise draws,

$$E[Q^{(k)}] = \text{tr}(C_{\text{post}}^{-1} V_{\text{post}}) + \mathbf{s}_b^\top C_{\text{post}}^{-1} \mathbf{s}_b.$$

The first (trace) term is the same regardless of how well the fixed truth matches the prior, since it does not involve $\mathbf{s}_{\text{true}}$ at all. Whether $E[Q^{(k)}]$ is close to, below, or far above p (the mean of χ_p^2) is therefore governed entirely by the bias term $\mathbf{s}_b^\top C_{\text{post}}^{-1} \mathbf{s}_b$: small for a well-matched truth, and potentially many orders of magnitude larger than p for a poorly matched one. Neither case should be read against the χ_p^2 benchmark from Experiment II, which relies specifically on $\mathbf{s}_{\text{true}}$ being a draw from the prior — a well-matched fixed truth does not restore that guarantee, it merely keeps the bias term small.

Two fixed truths, not one

This experiment should be run **twice**, with two qualitatively different deterministic truths, so that the comparison is informative rather than a foregone conclusion:

1. **A smooth truth**, e.g. the Gaussian-anomaly field from Experiment I (Step 4). This field is close to a typical draw from the assumed prior, so the bias term above is expected to be small and empirical coverage close to nominal — but, as derived above, this does *not* imply $Q^{(k)}$ close to χ_p^2 , since the truth is fixed rather than drawn from the prior.
2. **A sharp or discontinuous truth**, e.g. a checkerboard pattern or a rectangular block anomaly with hard edges. This field is atypical under a smooth squared-exponential prior, so the bias term above is expected to be large, giving poor empirical coverage and $Q^{(k)}$ far above χ_p^2 's mean, even though the posterior is mathematically well defined.

Running both cases side by side demonstrates the point of the experiment directly: a posterior can be exactly correct in the generative sense (Experiment II) while being a poor description of uncertainty for a specific physical truth the prior does not represent well (Experiment III, sharp case). This distinction — Bayesian calibration under the generative model versus performance on a fixed physical truth — is the central conceptual focus of the project.

10 Performance metrics

The project will use several complementary metrics.

Reconstruction error

$$E_{\text{rel}} = \frac{\|\mathbf{s}_{\text{post}} - \mathbf{s}_{\text{true}}\|_2}{\|\mathbf{s}_{\text{true}}\|_2}.$$

Posterior uncertainty

For each cell, $\sigma_{\text{post},j} = \sqrt{(C_{\text{post}})_{jj}}$. These values provide a spatial map of posterior uncertainty.

Standardized error

$$z_j = \frac{s_{\text{true},j} - s_{\text{post},j}}{\sigma_{\text{post},j}}.$$

For a correctly specified model, these quantities should exhibit the behavior predicted by the Gaussian posterior.

Mahalanobis error

$$Q = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})^\top C_{\text{post}}^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}).$$

Under the correctly specified Bayesian generative experiment, $Q \sim \chi_p^2$.

Credible-region coverage

For a nominal level $1 - \alpha$, construct marginal credible intervals and measure their empirical coverage over repeated synthetic experiments. The marginal interval for cell j is

$$s_{\text{post},j} \pm z_{1-\alpha/2} \sqrt{(C_{\text{post}})_{jj}},$$

where $z_{1-\alpha/2}$ is the corresponding standard-normal quantile. The empirical fraction of experiments for which $s_{\text{true},j}$ lies inside this interval can be compared with the nominal coverage level.

11 Numerical implementation

The implementation should be divided into independent components:

1. **Grid construction:** create the domain and cell centers.
2. **Ray construction:** generate source–receiver pairs and the corresponding straight line segments.

3. **Ray–cell intersection:** compute A_{ij} for every ray and cell.
4. **Prior construction:** construct the squared-exponential covariance C_s , including the numerical jitter term.
5. **Synthetic data generation:** generate $\mathbf{s}_{\text{true}}$ (both smooth and sharp variants), noiseless travel times, and noisy observations.
6. **Bayesian inversion:** compute $\mathbf{s}_{\text{post}}$ and C_{post} via Cholesky factorization and triangular solves.
7. **Diagnostics:** calculate reconstruction errors, posterior uncertainties, standardized errors, and Mahalanobis distances.
8. **Visualization:** produce maps of the true field, posterior mean, posterior standard deviation, and reconstruction error.
9. **Repeated experiments:** repeat the stochastic experiment over many independent noise/truth realizations.

12 Minimal experimental schedule

To keep the project computationally small and reproducible, the initial study will use the following sequence:

1. Implement a 20×20 grid.
2. Use a modest set of uniformly distributed sources and receivers.
3. Construct and visualize A .
4. Verify ray lengths using row sums of A .
5. Define a smooth deterministic anomaly.
6. Generate one noisy data set.
7. Compute the posterior.
8. Produce the three primary images: true field, posterior mean, posterior standard deviation.
9. Repeat the correctly specified Bayesian experiment (Experiment II) for approximately 100 independent realizations.
10. Plot the empirical distribution of the Mahalanobis statistic against the χ_p^2 reference distribution.
11. Repeat with a deterministic fixed truth (Experiment III), using *both* the smooth anomaly and a sharp/checkerboard truth.

13 Expected outputs

The project should produce the following concrete outputs:

1. A reproducible construction of the tomography sensitivity matrix A , validated against known ray lengths.
2. A numerical implementation of the exact Gaussian posterior.
3. A reconstruction of a synthetic slowness field from noisy travel-time measurements.
4. Spatial maps of posterior uncertainty.
5. Empirical calibration results under the correctly specified Bayesian model.
6. A comparison between the correctly specified Bayesian experiment and the fixed-truth experiment, for both a smooth and a sharp fixed truth.
7. A concluding discussion of what the fixed-truth comparison reveals about the relationship between prior assumptions, reconstruction quality, and uncertainty calibration.

14 Scope and interpretation

This project is intentionally a controlled linear model rather than a full seismic inversion. The following effects are excluded from the study: curved rays; nonlinear dependence of ray paths on the slowness field; attenuation; anisotropy; frequency-dependent effects; wave-equation simulation; non-Gaussian likelihoods.

The study is also scoped to a single acquisition geometry and a single, fixed prior/noise specification for the inference model: it does not include a systematic sweep over acquisition geometry, a controlled prior-misspecification experiment (varying the correlation length used to generate the truth against the one used for inference), or a controlled noise-misspecification experiment (varying the true versus assumed noise level). The fixed-truth comparison in Experiment III already demonstrates the central phenomenon these further experiments would also illustrate — that a mathematically well-defined posterior need not be a trustworthy description of physical uncertainty when the assumed model does not match the true generative process — via the mismatch between the sharp fixed truth and the smooth prior, rather than via a mismatch in the prior’s or the noise model’s parameters.

The purpose of these simplifications is to isolate the relationship between linear tomography, Bayesian regularization, and uncertainty quantification. The resulting model is sufficiently simple that the posterior can be computed exactly. This makes it possible to distinguish errors caused by numerical implementation from errors caused by statistical misspecification.

15 Final objective

The final objective is not simply to demonstrate that Bayesian linear tomography can reconstruct a synthetic image. Rather, the project should demonstrate the complete chain

geometry $\rightarrow A \rightarrow$ synthetic observations $\rightarrow$ Bayesian inversion $\rightarrow$ uncertainty $\rightarrow$ calibration in a controlled setting. The central comparison is between a *correctly specified Bayesian model* and a *fixed physical truth, either well matched to the prior’s assumed smoothness or deliberately*

chosen to be poorly represented by it. This provides a compact computational investigation of when a posterior distribution is not merely mathematically well defined, but also a useful quantitative description of uncertainty in a tomography problem.